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Prepare Physical Machines

Summary

To install Clan on physical machines, you need to start our custom installer image from a boot device. The minimal system will then be reachable by your setup machine and a deployment can be triggered remotely.

If you only plan to use virtual machines, you can skip this step.

[Why Nixos-Anywhere Doesn't Work On Physical Hardware?](#) >

[Reasons For A Custom Install Image](#) >

Requirements

- Estimated time for this step: 20 minutes
- A USB drive with at least 1.5GB total space (!! all data will be lost !!)
- A Linux/NixOS machine with internet access to create the boot stick. You can use your setup device or any other machine for this step.
- One or more physical target machines (!! all data will be lost !!)

Minimum target system requirements: 2 CPUs, 4GB RAM, 30gb HDD space, network interface

Identify the USB Flash Drive

1. Insert your USB flash drive into the Linux computer you want to create the boot stick on.
2. Identify your flash drive with `lsblk`:

```
lsblk
```

```

1 NAME                                MAJ:MIN RM   SIZE RO TYPE
2 sdb                                8:0    1 117,2G  0 disk
3 └─sdb1                            8:1    1 117,2G  0 part
4 nvme0n1                          259:0    0    1,8T  0 disk
5 └─nvme0n1p1                      259:1    0    512M  0 part
6 └─nvme0n1p2                      259:2    0    1,8T  0 part
7   └─luks-f7600028-9d83-4967-84bc-dd2f498bc486 254:0    0    1,8T  0 part

```

i In This Case The USB Device Is

3. Ensure all partitions on the drive are unmounted. Replace `sdb1` in the command below with your device identifier (like `sdc1`, etc.):

```
sudo umount /dev/sdb1
```

Installer Creation

Linux OS

Other OS

Create a Custom Installer

We recommend to build your own installer because of the following reasons:

- Include your ssh public keys into the image that allows passwordless ssh connection later on.
- Set your preferred language and keymap

```

1 clan flash write --flake https://git.clan.lol/clan/clan-core/archiv
2   --ssh-pubkey $HOME/.ssh/id_ed25519.pub \
3   --keymap us \
4   --language en_US.UTF-8 \
5   --disk main /dev/sd<X> \

```

6 flash-installer

Note

Replace `$HOME/.ssh/id_ed25519.pub` with a path to your SSH public key. Replace `/dev/sd<X>` with the drive path you want to flash

Specifying The Wrong Device Can Lead To Unrecoverable Data Loss.

The `clan flash` utility will erase the disk. Make sure to specify the correct device

- **SSH-Pubkey Option**

To add an ssh public key into the installer image append the option:

```
--ssh-pubkey <pubkey_path>
```

If you do not have an ssh key yet, you can generate one with `ssh-keygen -t ed25519` command. This ssh key will be installed into the root user.

- **Connect to the installer**

On boot, the installer will display on-screen the IP address it received from the network. If you need to configure Wi-Fi first, refer to the next section. If Multicast-DNS (Avahi) is enabled on your own machine, you can also access the installer using the `flash-installer.local` address.

- **List Keymaps**

You can get a list of all keymaps with the following command:

```
clan flash list keymaps
```

- **List Languages**

You can get a list of all languages with the following command:

```
clan flash list languages
```

Checkpoint 1: Boot From USB Stick on Target Device

To see if your new Clan USB boot stick works, plug it into a target device and boot from the USB drive with **secure boot turned off**.

For step by step instructions go to [Disabling Secure Boot](#)

(Optional) Connect to Wi-Fi Manually

If you don't have access via LAN, the installer offers support for connecting via Wi-Fi.

```
iwctl
```

This will enter

```
[iwd]#
```

Now run the following command to connect to your Wi-Fi:

```
1 # Identify your network device.
2 device list
3
4 # Replace 'wlan0' with your wireless device name
5 # Find your Wi-Fi SSID.
6 station wlan0 scan
7 station wlan0 get-networks
8
9 # Replace your_ssid with the Wi-Fi SSID
10 # Connect to your network.
11 station wlan0 connect your_ssid
12
13 # Verify you are connected
14 station wlan0 show
```

If the connection was successful, you should see something like this:

```
State                connected
Connected network    FRITZ!Box (Your router device)
```

IPv4 address 192.168.188.50 (Your new local ip)

Press ++ctrl+d++ to exit .

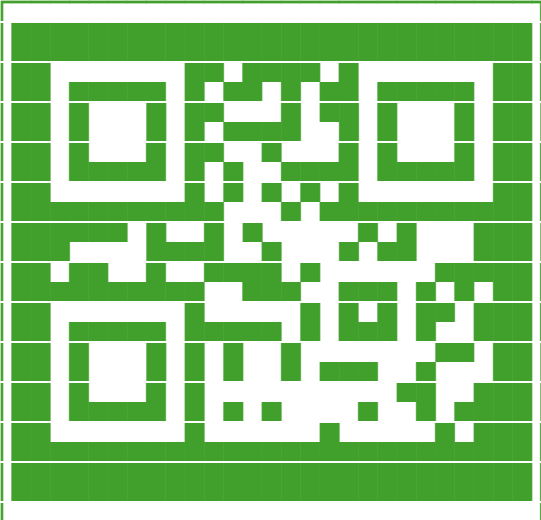
⚡ Important

Press ++ctrl+d++ **again** to update the displayed QR code and connection information.

Image Installer

The installer will randomly generate a password and local addresses on boot, then run a SSH server with these preconfigured. The installer shows its deployment relevant information in two formats, a text form, as well as a QR code. This is an example of the booted installer.

```
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```



```
# This is the QR Code (1)

Root password: cheesy-capital-unwell # password (2)
Local network addresses:
enp1s0          UP    192.168.178.169/24 metric 1024 fe80::21e:6ff:fe
enp2s0          DOWN
wlan0           DOWN # connect to wlan (3)
Onion address: 6evxy5yhzytwpnhc2vpscrbti3iktxdhpnf6yim6bbs25p4v6beemzy
Multicast DNS: nixos-installer.local

Press 'Ctrl-C' for console access
```

1. This is not an actual QR code, because it is displayed rather poorly on text sites. This would be the actual content of this specific QR code prettified:

```
1 {
2   "pass": "cheesy-capital-unwell",
3   "tor": "6evxy5yhzytwpnhc2vpscrbiti3iktxdhpnf6yim6bbs25p4v6beemzy",
4   "addrs": [
5     "2001:9e8:347:ca00:21e:6ff:fe45:3c92"
6   ]
7 }
```

To generate the actual QR code, that would be displayed use:

```
echo '{"pass":"cheesy-capital-unwell","tor":"6evxy5yhzytwpnhc2vpscrbiti3iktxdhpnf6yim6bbs25p4v6beemzy"}'
```

2. The root password for the installer medium. This password is autogenerated and meant to be easily typeable.

♥ Tip

For easy sharing of deployment information via QR code, we highly recommend using [KDE Connect](#).

There are two ways to deploy your machine:

Generating a Hardware Report

The following command will generate a hardware report with [nixos-facter](#) and writes it back into your machine folder. The `--phases kexec` flag makes sure we are not yet formatting anything, instead if the target system is not a NixOS machine it will use [kexec](#) to switch to a NixOS kernel.

Password

QR Code

Password

```
clan machines install [MACHINE] \
--update-hardware-config nixos-facter \
```

```
--phases kexec \
```

```
--target-host root@192.168.178.169
```

If you are using our template `[MACHINE]` would be `jon`

Up Next

If all your machines are physical, you can continue with disk preparations here:

[Next Step \(Choose Disk Format\)](#)

Alternatively, feel free to also add virtual machines into the mix during the next step!

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Prepare Virtual Machines

Need help? Reach out to [our community](#).